MTH 346/646 - Class 18 - 10/28

Fun fact: Let $F = \mathbb{C}(x, y)$ be the field of rational functions in x and y with complex coordinates. Consider the set of solutions to

$$u^3 + v^3 = x^3 + y^3$$

with $u, v \in F$. There are some obvious solutions like $(u, v) = (x, y), (y, x), (\zeta_3 x, y)$.

• There is a group law on the set of solutions coming from the fact that this is isomorphic to an elliptic curve. For example, one solution is

$$(u, v) = \left(\frac{x(x^3 + 2y^3)}{x^3 - y^3}, \frac{y(y^3 + 2x^3)}{y^3 - x^3}\right).$$

(This identity was written down by François Viète in 1591.)

• There is also a way to compose solutions, since if $(f_1(x, y), g_1(x, y))$ and $(f_2(x, y), g_2(x, y))$ are solutions, then

$$f_1(f_2(x, y), g_2(x, y))^3 + g_1(f_2(x, y), g_2(x, y))^3 = f_2(x, y)^3 + g_2(x, y)^3 = x^3 + y^3.$$

The set of solutions is isomorphic as a group of $\mathbb{Z} \times \mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$. On an index 3 subgroup, the "composition" operation gives a multiplication operation that turns the set of solutions into a ring.

Recall Lemma 4: The index $[C(\mathbb{Q}) : 2C(\mathbb{Q})]$ is finite.

If $C : y^2 = x^3 + ax^2 + bx$, define $\overline{C} : y^2 = x^3 + \overline{a}x^2 + \overline{b}x$ where $\overline{a} = -2a$ and $\overline{b} = a^2 - 4b$.

There is a function

$$\phi : C \to \overline{C}$$

given by

$$\phi((x, y)) = \left(\frac{y^2}{x^2}, \frac{y(x^2 - b)}{x^2}\right)$$

if $(x, y) \neq \mathcal{O}, T$ and $\phi(\mathcal{O}) = \phi(T) = \overline{\mathcal{O}}$. There is also a map $\psi : \overline{C} \to C$ that is similar.

Last time we proved that ϕ is a homomorphism, and $\psi \circ \phi(P) = 2P$ for all $P \in C(\mathbb{Q})$. The same arguments show that ψ is a homomorphism and $\phi \circ \psi(\overline{P}) = 2\overline{P}$ for $\overline{P} \in \overline{C}(\mathbb{Q})$.

Finishing Lemma 4

We will follow three steps to complete the proof of Lemma 4:

We have a homomorphism $\phi : C(\mathbb{Q}) \to \overline{C}(\mathbb{Q}) = \overline{C}(\mathbb{Q})$.

Step 1 - Understand the image of ϕ. We will show that

(a) $\mathcal{O}$ is in the image of ϕ,

(b) $\overline{T}$ is in the image of ϕ if and only if $\overline{b}$ is a perfect square, and

(c) $(\overline{x}, \overline{y})$ is in the image of ϕ if and only if $\overline{x}$ is a square.

Step 2 - Construct a homomorphism $\alpha : C(\mathbb{Q}) \to \mathbb{Q}^\times/(\mathbb{Q}^\times)^2$ so that $\ker \alpha = \operatorname{im} \psi$, and use this to conclude that $[C(\mathbb{Q}) : \psi(\overline{C}(\mathbb{Q}))]$ and $[\overline{C}(\mathbb{Q}) : \phi(C(\mathbb{Q}))]$ are both finite.

Step 3 - Use this to prove that $[C(\mathbb{Q}) : 2C(\mathbb{Q})]$ is finite.

Step 1 - Characterize the image of ϕ

(a) We have that $\phi(\mathcal{O}) = \phi(T) = \overline{\mathcal{O}}$.

(b) Since $\phi(x, y) = \left(\frac{y^2}{x^2}, \frac{y(x^2-b)}{x^2}\right)$, we have that $\phi(x, y) = \overline{T}$ implies that $x^2 \neq 0$ (since $x = 0$ forces $(x, y) = T$ and so $\phi(T) = \overline{\mathcal{O}}$). Thus, $y^2 = 0$ but $x \neq 0$. This means that there is a rational point $(x, 0)$ with $x \neq 0$ on $C : y^2 = x^3 + ax^2 + bx$ (or, in other words, all three points of order 2 on C are rational).

We get $0 = x(x^2 + ax + b)$, and the quadratic $x^2 + ax + b$ must have a rational root. The quadratic formula implies that $a^2 - 4b = \overline{b}$ must be a square.

(c) It is clear that the x-coordinate of $\phi(x, y)$ is a square. Conversely, suppose that $(\overline{x}, \overline{y}) \in \overline{C}(\mathbb{Q})$ and that $\overline{x} = w^2$ is a perfect square (that is, $w \in \mathbb{Q}$). We will construct $(x, y) \in C(\mathbb{Q})$ so that $\phi(x, y) = (\overline{x}, \overline{y})$. The case that $w = 0$ was handled in (b). Assume that $w \neq 0$.

Let $P_1 = (x_1, y_1)$ and $P_2 = (x_2, y_2)$, where

$$x_i = \frac{1}{2}\left(w^2 - a \pm \frac{\overline{y}}{w}\right), y_i = (\pm w)x_i.$$

(We take the plus in all formulas if $i = 1$ and the minus if $i = 2$.) We have that

$$\begin{aligned} x_1x_2 &= \frac{1}{4}\left((w^2 - a)^2 - \frac{\overline{y}^2}{w^2}\right) \\ &= \frac{1}{4}\left((\overline{x} - a)^2 - \frac{\overline{y}^2}{\overline{x}}\right) \\ &= \frac{1}{4}\left(\overline{x}^2 - 2a\overline{x} + a^2 - (\overline{x}^2 + \overline{a}\overline{x} + \overline{b})\right) \\ &= \frac{1}{4}(a^2 - \overline{b}) = \frac{1}{4}(a^2 - (a^2 - 4b)) = b. \end{aligned}$$

(Recall that $\overline{a} = -2a$ so $-2a\overline{x} - \overline{a}\overline{x} = 0$.)

To show that P_1 and P_2 are on C, we need to know that

$$\begin{aligned} w^2x_i^2 &= x_i^3 + ax_i^2 + bx_i \\ w^2 &= x_i + a + \frac{b}{x_i} \\ &= x_1 + a + x_2. \end{aligned}$$

When we add x_1 and x_2 we get

$$x_1 + x_2 = w^2 - a$$

and so $x_1 + a + x_2 = w^2$. Thus, P_1 and P_2 are on C.

Now we must calculate $\phi(P_i)$. For the x-coordinate, we get

$$\frac{y_i^2}{x_i^2} = \frac{((\pm w)x_i)^2}{x_i^2} = w^2.$$

For the y-coordinate, we get

$$\begin{aligned}\frac{y_1(x_1^2 - b)}{x_1^2} &= \frac{y_1(x_1^2 - x_1x_2)}{x_1^2} \\ &= \frac{y_1(x_1 - x_2)}{x_1} \\ &= w(x_1 - x_2).\end{aligned}$$

A similar calculation shows that

$$\frac{y_2(x_1^2 - b)}{x_2^2} = w(x_1 - x_2).$$

Now, $w(x_1 - x_2) = w\left(\frac{\overline{y}}{w}\right) = \overline{y}$. Thus, the y-coordinates of $\phi(P_1)$ and $\phi(P_2)$ are both $\overline{y}$. This proves the desired statement.

The group $\mathbb{Q}^\times/(\mathbb{Q}^\times)^2$

The group $\mathbb{Q}^\times/(\mathbb{Q}^\times)^2$ is the quotient group of the multiplicative group of rational numbers by the subgroup of squares. One way to think of this is that for $c, d \in \mathbb{Q}^\times$, we say that

$$c \equiv d \pmod{(\mathbb{Q}^\times)^2}$$

if c/d is the square of a nonzero rational number. The binary operation on $\mathbb{Q}^\times/(\mathbb{Q}^\times)^2$ is multiplication.

For every $c \in \mathbb{Q}^\times$ there is a unique squarefree integer d (an integer not divisible by the square of any prime) so that $c(\mathbb{Q}^\times)^2 = d(\mathbb{Q}^\times)^2$, or equivalently $c \equiv d \pmod{(\mathbb{Q}^\times)^2}$.

Examples: Let $a = -3/5$ and $b = 5/7$. Then the unique squarefree integer d so that $ab \equiv d \pmod{(\mathbb{Q}^\times)^2}$ is -21.

Mod squares, we have

$$ab = (-3/5) \cdot (5/7) = -3/7 \equiv (-3/7) \cdot (7^2) \equiv -21 \pmod{(\mathbb{Q}^\times)^2}.$$

The group $(\mathbb{Q}^\times)/(\mathbb{Q}^\times)^2$ has the property that every element has order 1 or 2. The group $(\mathbb{Q}^\times)/(\mathbb{Q}^\times)^2$ is infinite (and it is not finitely generated).

Step 2 - The homomorphism α

The homomorphism $\alpha : C(\mathbb{Q}) \to \mathbb{Q}^\times/(\mathbb{Q}^\times)^2$ is defined by

$$\alpha(P) = \begin{cases} 1 & \text{if } P = \mathcal{O} \\ b & \text{if } P = T \\ x & \text{if } P = (x, y), x \neq 0. \end{cases}$$

Proposition: (a) The map $\alpha : C(\mathbb{Q}) \to \mathbb{Q}^\times/(\mathbb{Q}^\times)^2$ is a homomorphism.

(b) The kernel of α is the image of $\psi(\overline{C}(\mathbb{Q}))$. The first isomorphism theorem thus gives

$$C(\mathbb{Q})/\psi(\overline{C}(\mathbb{Q})) \cong \alpha(C(\mathbb{Q})) = \operatorname{im} \alpha.$$

(c) Let p_1, p_2, …, p_t be the distinct primes dividing b. Then the image of α is contained in the subgroup of $\mathbb{Q}^\times/(\mathbb{Q}^\times)^2$ consisting of the elements

$$\{\pm p_1^{\epsilon_1} p_2^{\epsilon_2} \cdots p_t^{\epsilon_t} : \epsilon_i = 0 \text{ or } 1\}.$$

(d) $[C(\mathbb{Q}) : \psi(\overline{C}(\mathbb{Q}))]$ is at most 2^{t+1}.

Proof:

(a) First, observe that $\alpha(-P) = \alpha((x, -y)) = x \equiv \frac{1}{x} = \alpha((x, y))^{-1} = \alpha(P)^{-1}$. Thus, α sends inverses to inverses. So, in order to prove that α is a homomorphism, we will assume that $P_1 + P_2 + P_3 = \mathcal{O}$ and prove that $\alpha(P_1)\alpha(P_2)\alpha(P_3) \equiv 1 \pmod{(\mathbb{Q}^\times)^2}$.

The curve is $y^2 = x^3 + ax^2 + bx$. Suppose that $P_1 = (x_1, y_1)$, $P_2 = (x_2, y_2)$ and $P_3 = (x_3, y_3)$ are all on the line $y = \lambda x + \nu$. Then, x_1, x_2 and x_3 are roots of

$$\begin{aligned} x^3 + ax^2 + bx - (\lambda x + \nu)^2 &= x^3 + (a - \lambda^2)x^2 + (b - 2\lambda\nu)x - \nu^2 \\ &= (x - x_1)(x - x_2)(x - x_3). \end{aligned}$$

Multiplying out the last expression and comparing constant terms gives $-x_1x_2x_3 = -\nu^2$ and so $x_1x_2x_3 = \alpha(P_1)\alpha(P_2)\alpha(P_3) = \nu^2 \equiv 1 \pmod{(\mathbb{Q})^2}$.

(We have to be careful. What if ν is zero?)

(b) The kernel of α is all points in $C(\mathbb{Q})$ whose x-coordinate is a square. The calculation in Step 1 shows that if a point in $C(\mathbb{Q})$ has a square x-coordinate, then it is in the image of $\psi : \overline{C}(\mathbb{Q}) \to C(\mathbb{Q})$.

(c) Suppose that $(x, y) \in C(\mathbb{Q})$ and write $x = m/e^2$ and $y = n/e^3$. Plugging this in to $y^2 = x^3 + ax^2 + bx$ gives

$$\frac{n^2}{e^6} = \frac{m^3}{e^6} + \frac{am^2}{e^4} + \frac{bm}{e^2}.$$

Clearing denominators gives

$$n^2 = m^3 + am^2e^2 + bme^4 = m(m^2 + ame^2 + be^4).$$

We have that m and $m^2 + ame^2 + be^4$ multiply to be a perfect square. Let $d = \gcd(m, m^2 + ame^2 + be^4)$.

If $d = 1$, then m and $m^2 + ame^2 + be^4$ are both $\pm$ a perfect square.

Since $d|m$, $d|m^2 + ame^2$ and hence $d|be^4$. However, $\gcd(m, e) = 1$ and so $d|b$.

Every prime factor of $m(m^2 + ame^2 + be^4)$ must occur to an even power.

The only prime factors that m and $m^2 + ame^2 + be^4$ have in common are the prime divisors of b.

Thus,

$$m = \pm(\text{perfect square}) \prod_{i=1}^{t} p_i^{\epsilon_i}$$

where $\epsilon_i \in \{0, 1\}$ and the p_i are the prime factors of b.

(d) By the first isomorphism theorem, we have that

$$\alpha(C(\mathbb{Q})) \cong C(\mathbb{Q})/(\ker \alpha).$$

We have that $\ker \alpha = \psi(\overline{C}(\mathbb{Q}))$ by Step 1, and so

$$\alpha(C(\mathbb{Q})) \cong C(\mathbb{Q})/\psi(\overline{C}(\mathbb{Q})).$$

Thus, $[C(\mathbb{Q}) : \psi(\overline{C}(\mathbb{Q}))] = |C(\mathbb{Q})/\psi(\overline{C}(\mathbb{Q}))| = |\alpha(C(\mathbb{Q}))| \leq 2^{t+1}$. This proves the proposition.

Example: Let $C : y^2 = x^3 + 10x^2 - 112x$. I will show that $C(\mathbb{Q})$ has rank at least 2.

It's not too hard to find the points $(0, 0)$, $(14, -56)$, $(-16, -16)$ and $(-2, 16)$ in $C(\mathbb{Q})$. Their images under α are

$$\begin{aligned}
\alpha((0,0)) &\equiv -112 \equiv -7 \pmod{(\mathbb{Q}^\times)^2} \\
\alpha((14,-56)) &\equiv 14 \pmod{(\mathbb{Q}^\times)^2} \\
\alpha((-16,-16)) &\equiv -16 \equiv -1 \pmod{(\mathbb{Q}^\times)^2} \\
\alpha((-2,16)) &\equiv -2 \pmod{(\mathbb{Q}^\times)^2}.
\end{aligned}$$

The subgroup of $\mathbb{Q}^\times/(\mathbb{Q}^\times)^2$ generated by these consists of $\{1, -1, 2, -2, 7, -7, 14, -14\}$ which has order 8. So $|\text{im } \alpha|$ is a multiple of 8.

On the other hand, if $C(\mathbb{Q}) \cong T \times \mathbb{Z}^r$, then the image of α can be generated by $\alpha(T)$ together with the images under α of the r different points of infinite order.

The torsion subgroup of $C(\mathbb{Q})$ is $\mathbb{Z}/2\mathbb{Z}$ and generated by $(0, 0)$. This means that the image has size at most $2 \cdot 2^r = 2^{r+1}$. So $8 \leq 2^{r+1}$ this implies that $r \geq 2$.

Later on, we will develop theory to determine the rank of $C(\mathbb{Q})$ knowing im α and im $\overline{\alpha}$.

(It turns out that in this example, $C(\mathbb{Q})$ has rank 3 and that im $\overline{\alpha}$ also contributes to the rank.)